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Table 1 Levels of Several Cutin Precursor Classes Found in Wild-Type and *dso*-Mutant *Arabidopsis thaliana* Cuticles

Cutin precursor class	Level in wild-type cuticles (ng/cm^2)	Level in *dso*-mutant cuticles (ng/cm^2)
Alkanoic acids	25.0	12.5
2-hydroxy acids	80.0	20.0
ω-hydroxy acids	7.5	1.5
α,ω-dicarboxylic acids	60.0	10.0

Defective DSO protein in the leaves of *Arabidopsis thaliana* causes the greatest percent decrease in levels of extracellular cutins that contain which class of precursors?

- A. Alkanoic acids [4%]
- B. 2-hydroxy acids [16%]
- C. ω-hydroxy acids [15%]
- ✔ D. α,ω-dicarboxylic acids [62%]

Correct
61% answered correctly

Explanation:

微信Mai_ange

Percent decrease in mutants relative to wild-type

Cutin precursor class	Level in wild-type cuticles (ng/cm^2)	Level in *dso*-mutant cuticles (ng/cm^2)	% decrease
Alkanoic acids	25.0	12.5	50%
2-hydroxy acids	80.0	20.0	75%
ω-hydroxy acids	7.5	1.5	80%
α,ω-dicarboxylic acids	60.0	10.0	83.3%

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Table 1 shows the levels of several types of cutin precursors found in the cuticles of wild-type and *dso*-mutant plants. The passage states that these precursors were obtained by hydrolyzing cutin molecules into their precursor components. Therefore, the relative compositions of the cutins present in the plant cuticles can be inferred. All cutin precursor levels were reduced in the cuticles of *dso*-mutant plants relative to their levels in wild-type plants, with some being reduced to a greater extent than others.

The question asks for the type of cutin precursor that is reduced by the greatest percentage when DSO is defective. According to Table 1, α,ω-dicarboxylic acids are reduced from 60.0 ng/cm^2 in wild-type plants to 10.0 ng/cm^2 in *dso*-mutant plants. Therefore, these precursors are reduced to one sixth (16.7%) of their wild-type level when DSO is defective. In other words, defective DSO caused an **83.3% reduction** in extracellular **α,ω-dicarboxylic acid** levels. This is the **largest percent decrease** of the cutin precursors measured.

(Choice A) Alkanoic acid levels were decreased by 50% from 25.0 ng/cm^2 to 12.5 ng/cm^2.

(Choice B) Although 2-hydroxy acids underwent the greatest *total* decrease (60.0 ng/cm^2) when DSO was defective, the question asks for *percent* decrease. The decrease from 80.0 ng/cm^2 to 20.0 ng/cm^2 represents a 75% decrease, which is less than the 83.3% decrease undergone by α,ω-dicarboxylic acids.

(Choice C) ω-hydroxy acids decreased from 7.5 ng/cm^2 to 1.5 ng/cm^2, which is an 80% decrease.

Educational objective:
Experimental results are often assessed as a percentage of baseline levels to determine the relative effect of an experimental condition on the outcome.

Time spent:0 sec
Subject:
Biochemistry

QID:403294
System:
Carbohydrates, Nucleotides, and Lipids

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The cells of plants are coated with an outer layer called a cuticle that helps protect against water loss, frost, ultraviolet radiation, and damage from insects. The cuticle primarily consists of a molecule called cutin, which is composed of wax molecules linked together. These wax precursors of cutins are produced and linked together in the cytoplasm to form functional cutin, which is then exported to the extracellular matrix.

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α,ω-dicarboxylic acids	60.0	10.0

Based on the hypothesized localization of DSO proteins, the region of the plasma membrane that surrounds DSO is most likely rich in which of the following molecules relative to other plasma membrane regions?

- A. Sterols [39%]
- B. Phospholipids [39%]
- C. Prostaglandins [5%]
- D. Triglycerides [15%]

Correct
39% answered correctly

Explanation:

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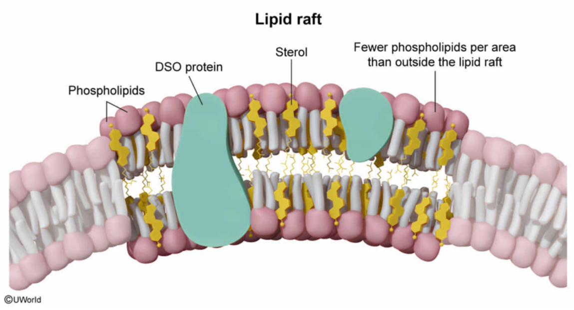

Lipid rafts are regions of relatively high order (ie, low entropy) of the cell membrane. These regions are believed to help localize distinct processes or pathways to specific regions of the membrane by recruiting the proteins and other components involved and holding them in proximity to each other (rather than being allowed to drift freely). To maintain a more ordered state, lipid rafts tend to be rich in sterols (cholesterol in animals and similar molecules called phytosterols in plants) and relatively poor in phospholipids compared to other regions of the membrane.

The passage states that DSO is hypothesized to be localized to lipid rafts. Therefore, DSO is most likely in an environment that is **rich in cholesterol** compared to other regions of the plasma membrane.

(Choice B) Lipid rafts have a lower percentage of phospholipids than nonraft regions of the membrane, so DSO is expected to be in a region that is relatively poor in phospholipids.

(Choice C) Prostaglandins are nonhydrolyzable lipids that are involved in inflammatory responses. They are not generally found in cell membranes.

(Choice D) Triglycerides are molecules in which three fatty acids are each linked to a glycerol molecule. These molecules are used for energy storage rather than structure, and are not generally found in cell membranes.

Educational objective:
Lipid rafts are relatively ordered domains within a cell membrane that help localize the components of certain processes to a defined region. These domains tend to be rich in cholesterol and relatively poor in phospholipids.

Time spent:0 sec
Subject: Biochemistry

QID:403295
System: Carbohydrates, Nucleotides, and Lipids

微信Mai_ange

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Cutin precursor class	Level in wild-type cuticles (ng/cm^2)	Level in *dso*-mutant cuticles (ng/cm^2)
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2-hydroxy acids	80.0	20.0
ω-hydroxy acids	7.5	1.5
α,ω-dicarboxylic acids	60.0	10.0

Which additional observation would support the conclusion that the DSO protein transports cutins to the extracellular matrix?

A. Wild-type plants synthesize more cutin precursors in a given amount of time than *dso*-mutant plants. [9%]

B. Wild-type plants degrade extracellular cutin at a lower rate than *dso*-mutant plants. [6%]

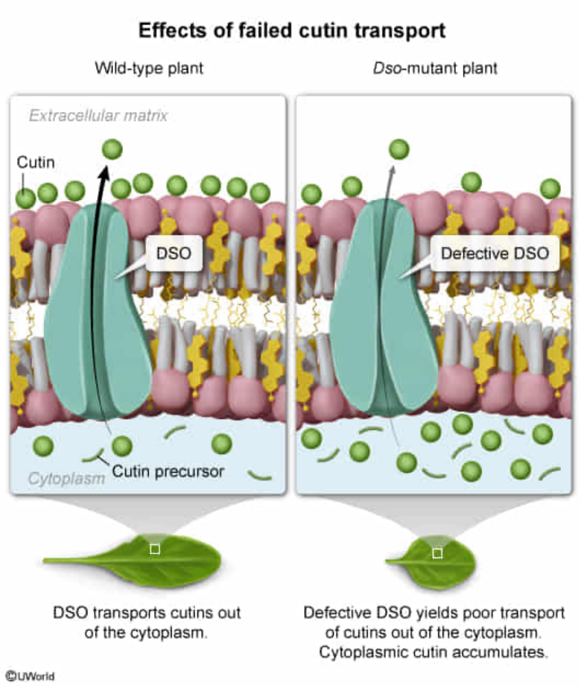
C. *dso*-mutant plants have high levels of cutin and cutin precursors in the cytosol relative to wild-type plants. [70%]

D. *dso*-mutant plants assemble cutin precursors into functional cutin less efficiently than wild-type plants. [13%]

Correct
71% answered correctly

Explanation:

微信Mai_ange



Experimental observations may have **more than one explanation**. For conclusions to be drawn from such experiments, **additional observations must be made** to either rule out or support each possible explanation.

The results presented in Table 1 indicate that cutin components are found at lower levels in the cuticles of *dso* mutant plants than in wild-type plants. The passage also states that cutins are assembled in the cytoplasm prior to transport to the extracellular matrix (ECM), where the cuticle is located. Therefore, one possible explanation for the *dso*-associated decrease of extracellular cutin is that DSO transports cutin from the cytoplasm to the ECM, and failed transport deprives the cuticle of cutin. However, it is *also* possible that DSO is involved in the *synthesis* of cutin precursors or the *assembly* of precursors into functional cutin, and that *dso*-mutant plants simply fail to produce cutin in the first place.

If DSO transports cutins, then cutin precursor synthesis and assembly into functional cutin may not be affected in *dso*-mutant plants. Consequently, if mutant plants exhibit an **abnormally high accumulation** of **cutin** and its **precursors in** the **cytoplasm** but less cutin in the cuticle as previously observed, **this new observation** would **support** the **conclusion** that **DSO transports cutins from** the **cytoplasm to** the **ECM**.

(Choice A) If wild-type plants synthesized more cutin precursors than *dso*-mutant plants, this would suggest that functional DSO is involved in precursor synthesis rather than transport.

(Choice B) If wild-type plants degraded extracellular cutin at a lower rate than *dso*-mutant plants, this would suggest that functional DSO inhibits cutin degradation.

(Choice D) If *dso*-mutant plants assembled cutin precursors into functional cutin less efficiently than wild-type plants, this would suggest that DSO plays a role in cutin assembly rather than transport.

Educational objective:

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Experimental observations may have multiple explanations. To support specific conclusions, additional experiments may be needed.

Time spent:0 sec	**QID**:403296
Subject:	**System:**
Biochemistry	Carbohydrates, Nucleotides, and Lipids

微信Mai_ange

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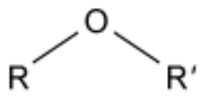
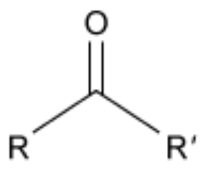
α,ω-dicarboxylic acids	60.0	10.0

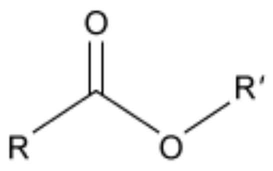
Which of the following functional groups is most likely abundant in cutin molecules?

A.


[12%]

B.


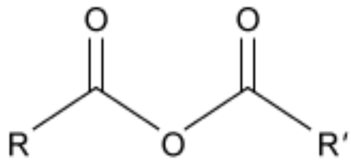
[9%]

C.


[58%]

D.

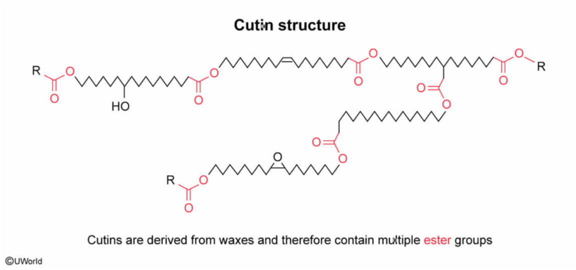
微信Mai_ange



[20%]

Correct
57% answered correctly

Explanation:



Waxes are lipid molecules that typically form soft solids that are malleable at room temperature. Most waxes are composed of two sets of long-chain hydrocarbons linked together through an ester group, which generally forms when a long-chain alcohol reacts with a long-chain fatty acid.

The passage states that cutins are composed of multiple wax molecules linked together; therefore, cutins most likely contain multiple ester groups. This class of functional group is correctly depicted by the diagram shown in Choice C.

(Choices A, B, and D) These diagrams depict an ether, a ketone, and an acid anhydride, respectively. Although these functional groups may be present in waxes, ester groups are much more common and more likely to be abundant in cutin.

Educational objective:
Most waxes in nature consist of long alkyl chains linked together through an ester bond.

Time spent:0 sec	**QID**:403297
Subject:	**System:**
Biochemistry	Carbohydrates, Nucleotides, and Lipids

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